

Tip: The Ionic Framework is open source and its source code is available on GitHub. Developers and programmers are encouraged to participate and contribute to the project.

Pre-requisites for building the app

Before starting to build your first Ionic app, there are few pre-requisites. You need to install a few of the following apps.

Node.js: Download and install Node.js (<https://nodejs.org/>). Ionic is also distributed as an npm package, so we will use Node and npm to install the Ionic Framework.

TypeScript: TypeScript is the superset programming language of JavaScript (we can think of it as a typed version of JavaScript) and we will code our app in it. After installing Node.js, using a terminal run the following command:

```
npm install -g typescript
```

Cordova: Cordova helps us to convert our HTML, CSS and JavaScript/TypeScript code into an app.

To install it, run the following command:

```
npm install -g cordova
```

Ionic: And now, install Ionic Framework as follows:

```
npm install -g ionic
```

Tip: You can also install these in one go by running the following command in the terminal/console:

```
npm install -g typescript cordova ionic
```

Note: It is assumed that you have some basic knowledge of Web technologies like HTML, CSS and JavaScript. If you don't, W3Schools (<http://www.w3schools.com/>) is a good place to start. The site has some great tutorials for Web technologies that are easy to follow.

Setting up the environment

Let's start. Type the following command in the command prompt console:

```
ionic start
```

You will be asked to name the project. Give it a name, e.g., Todo App, etc. You will then be prompted to pick a framework. In our case, we will pick Angular. Then we will be asked to select a template — we will select a blank one.

```
? Starter template: (Use arrow keys)
> tabs          | A starting project
  with a simple tabbed interface
  sidemenu      | A starting project
  with a side menu with navigation in the
  content area
  blank         | A blank starter
  project
  my-first-app | An example application
  that builds a camera with gallery
  conference   | A kitchen-sink
  application that shows off all Ionic
  has to offer
```

After the template is selected and built, go to your newly created project directory and run *ionic serve* within the app directory to see your app.

```
ionic serve
```

The command will start the local development server. Next, it will load <http://localhost:8100/home> in your browser. If you see the output screen shown in Figure 1, you have successfully installed all the dependencies required.

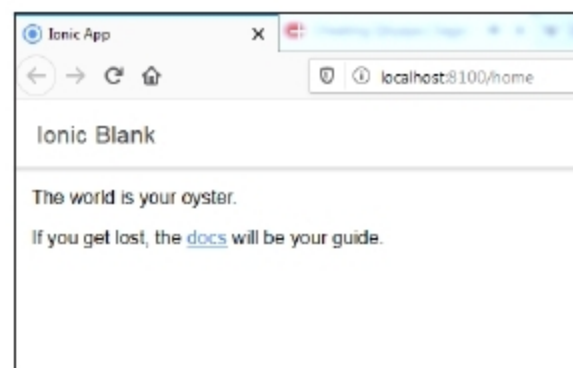


Figure 1: Ionic project running in the browser

Getting started

Let's have a look at the project file structure (see Figure 2).

As shown in Figure 2, the main source files consist of the following.

- *app.module.ts*: This is the entry point of the mobile app. All components, modules, pages, etc, must be added to this file.
- *app.component.ts*: This is the first logic that gets executed when the app starts running.
- *app.html*: This file contains the view/template of the app; other UI pages get mounted here.
- *app.scss*: This file contains all the CSS styling and SASS variables to be used globally within the app.

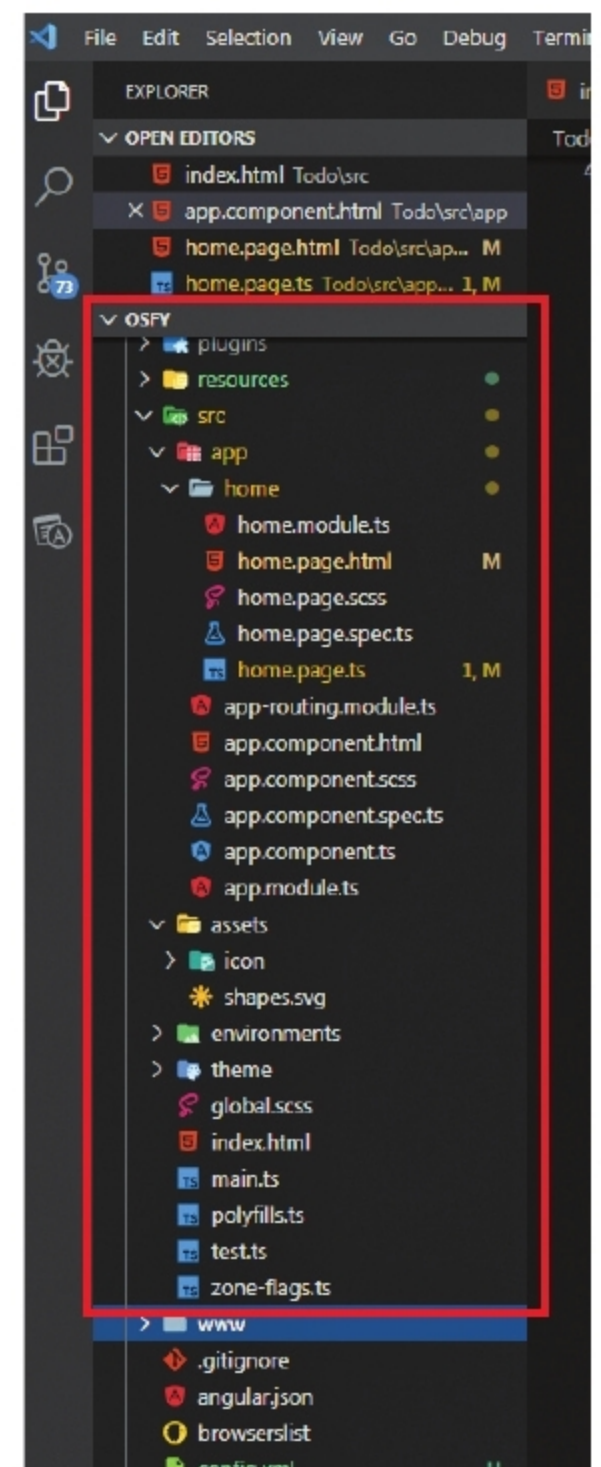


Figure 2: Project file structure